

Anti-static Materials for Labs

Various studies around the world reveal similar shocking facts, such as 60-90 per cent of devices are damaged due to electrostatic discharge (ESD), and 70 per cent of these failures are due to ungrounded people. So it becomes important that you take ESD control systems seriously. Otherwise, losses can be astonishingly high. You can follow simple ESD control systems using anti-static materials and achieve the highest reliability in the output



(Image courtesy: www.procat.com)

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Electrostatic discharge (ESD) is the rapid transfer of electrostatic charge between two objects. This normally happens when two objects at different potentials come into direct contact. This static charge buildup is majorly caused by tribocharging and electrostatic induction.

Tribocharging is the generation of static electricity due to separation of electrical charges that occurs when two mate-

rials are brought into contact and then separated. For example, when you walk on a carpet, the friction between your shoes and carpet results in tribocharging, thus creating a difference of electrical potential that can lead to an ESD event.



Fig. 1: Tribocharging (Image courtesy: ESD Systems)

Electrostatic induction occurs when an electrically charged object comes near a conductive object isolated from ground. The charged object produces an electrostatic field that causes electrical charges on the surface of the other object to redistribute as shown in Fig. 2. As such, the net charge on the object is zero but it has surfaces with excess positive and negative charge, so an ESD event can happen from these surfaces.



Fig. 2: Electrostatic Induction

How does ESD affect electronics

Electronic products are becoming smaller and faster but also becoming more sensitive to ESD. ESD can damage sensitive electronic components, resulting in failures, reduced reliability and increased rework costs. The worst part is that most often the chips get damaged only partially and start malfunctioning later. The cost of a single failure can be astonishing—in defence, such failures can cost lives.